

VON
**PHYSIK,
INFORMATIK**
UND DEM GANZEN REST



Alice



Alice

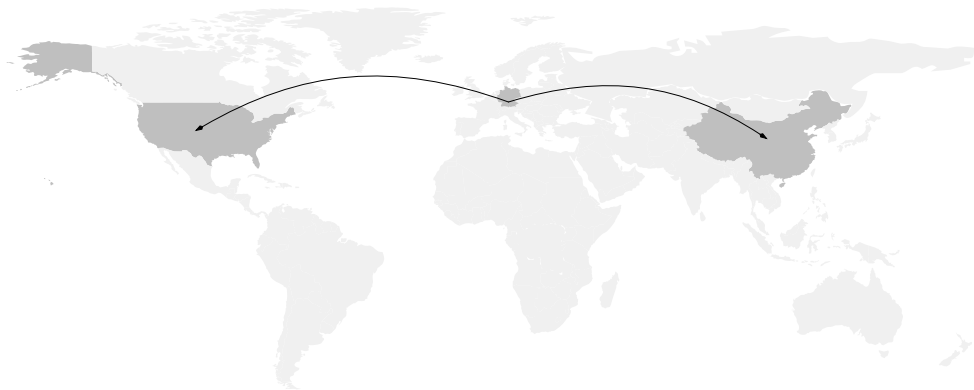
Hallo



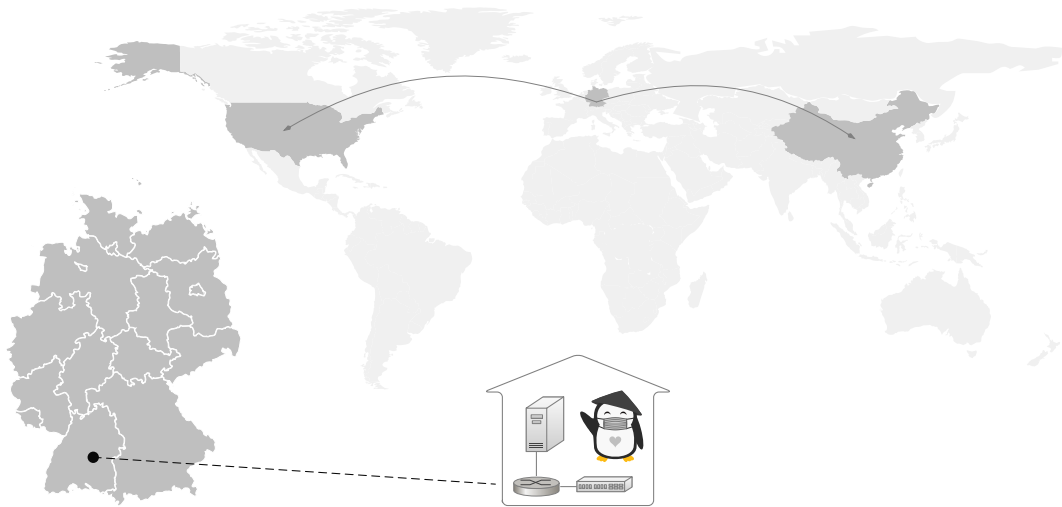
Bob

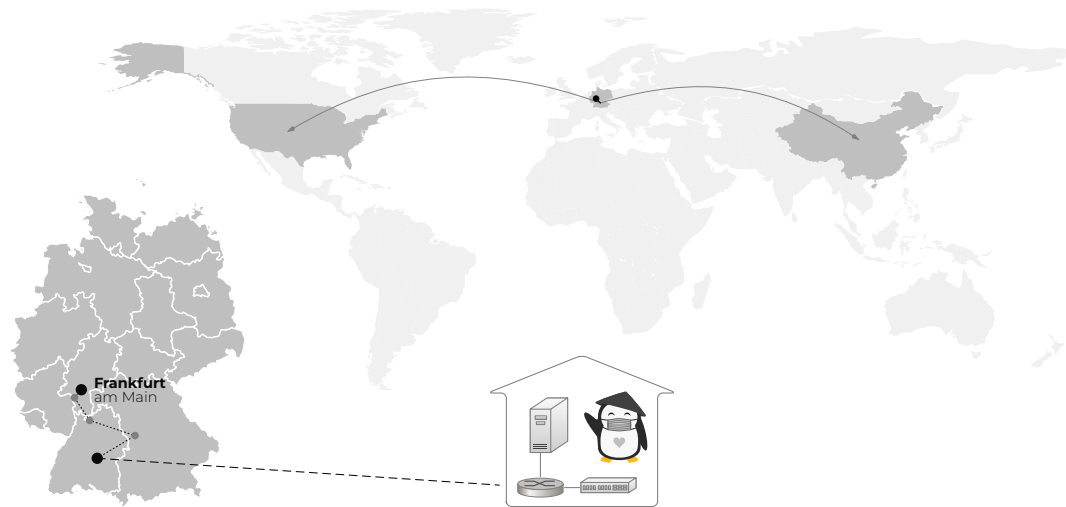


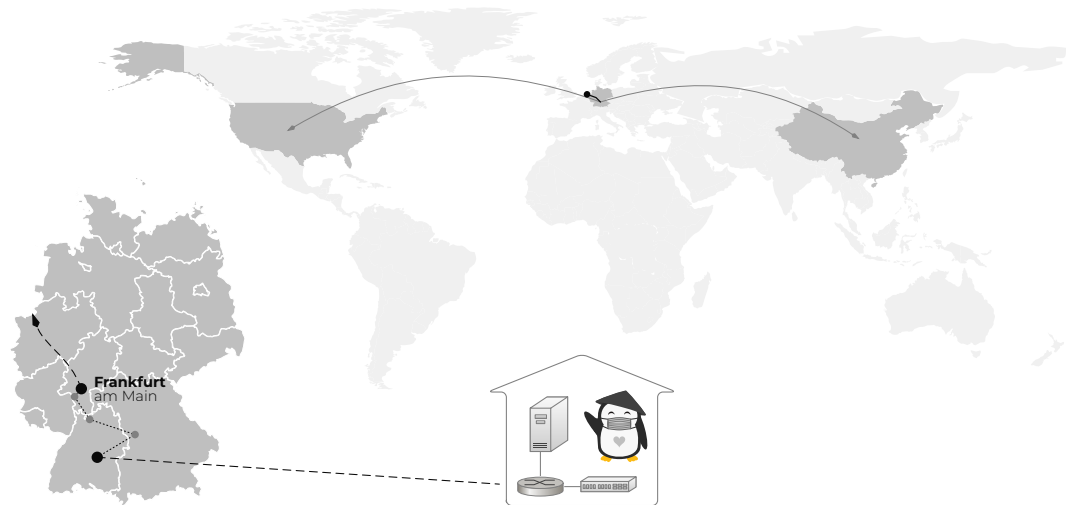


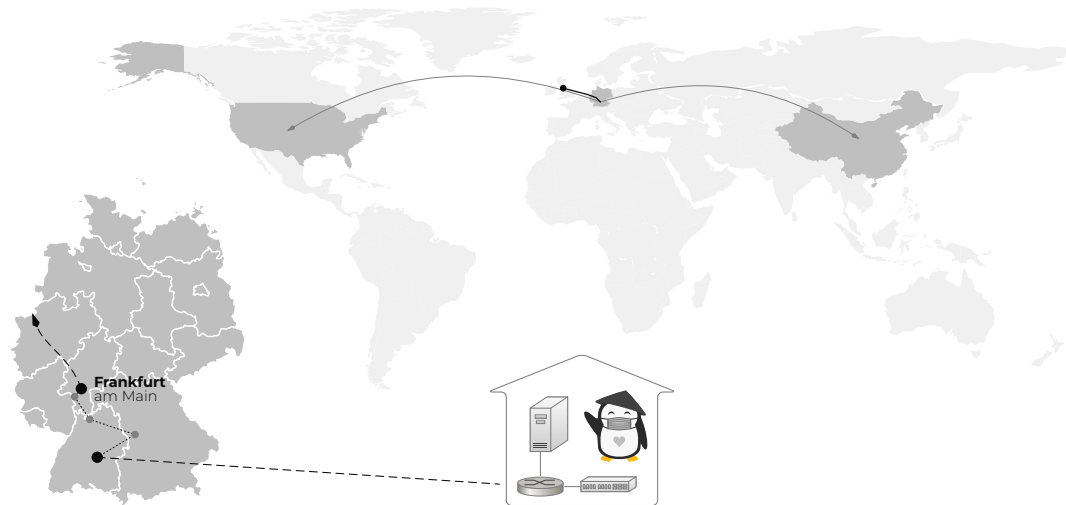


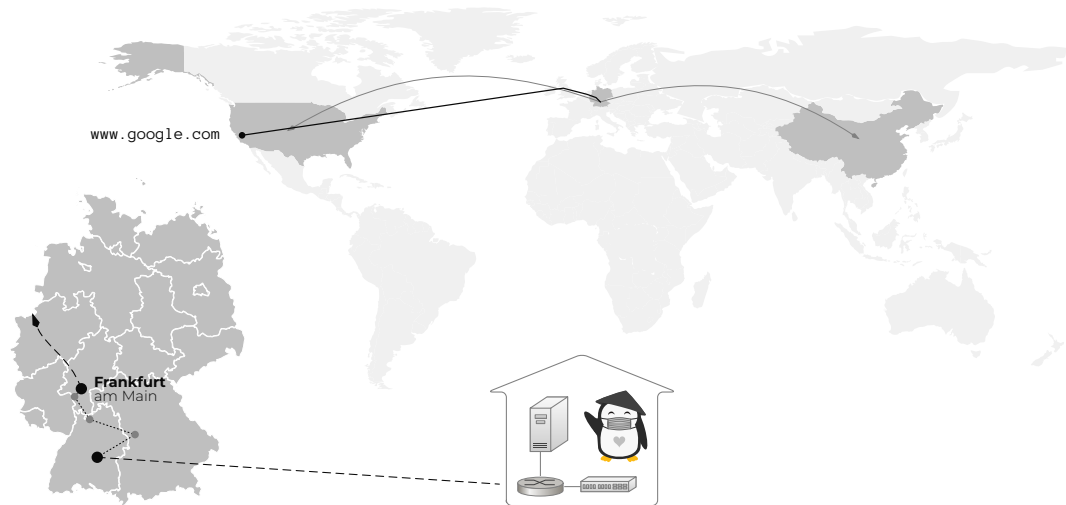


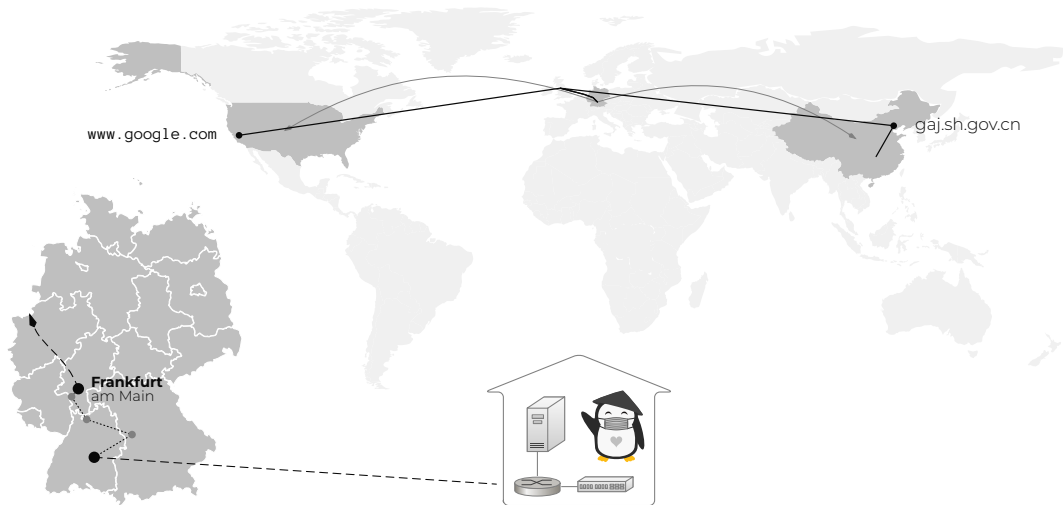












Network Edge

Network Core

Network Edge

application
transport
network
link
physical

Mobilnetzwerk



DSLAM

Tel. Net.

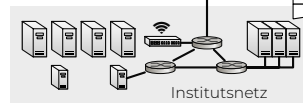
reg. ISP

glob. ISP

network
link
physical

header val.	output link
0100	3
0101	2
0111	2
1011	1
1111	4

application
transport
network
link
physical

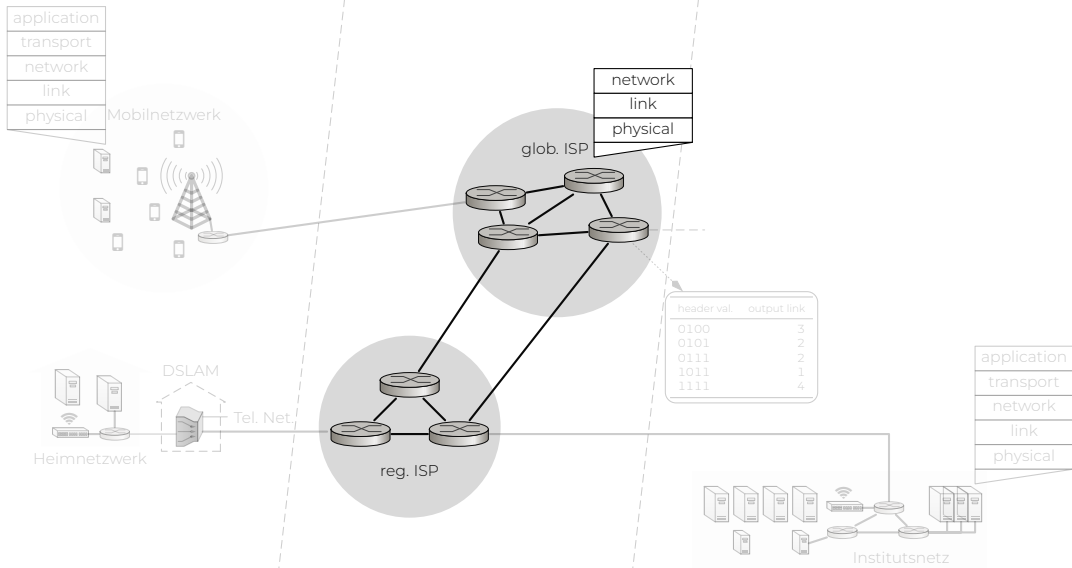


Institutsnetz

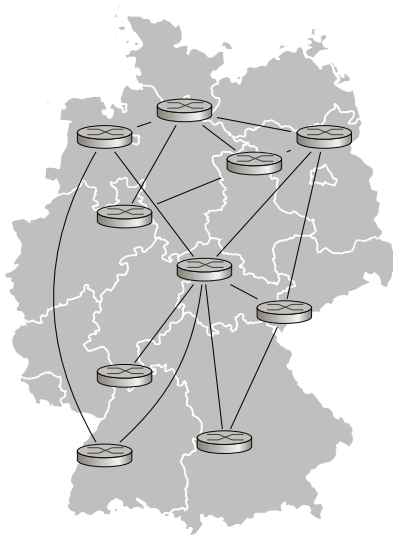
Network Edge

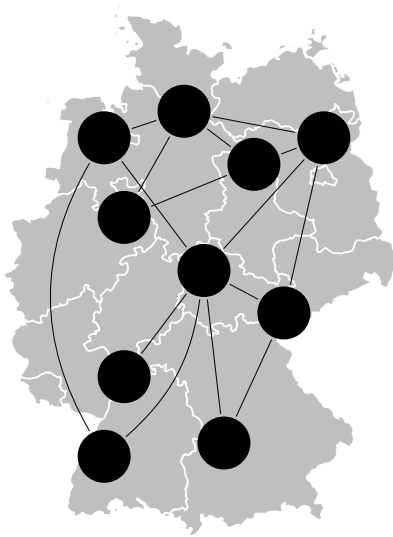
Network Core

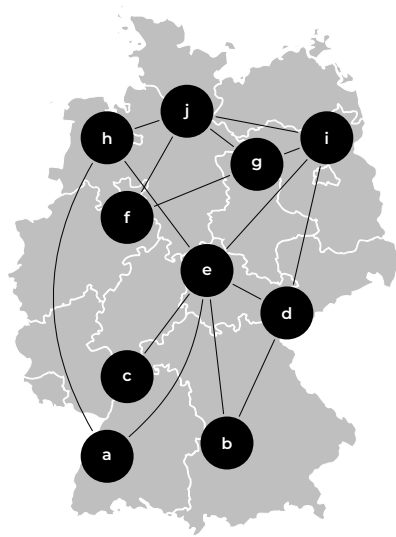
Network Edge

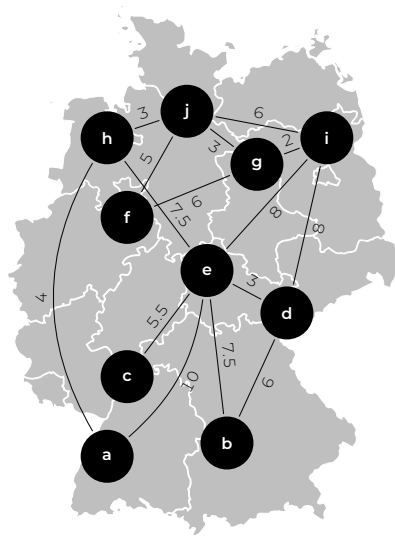


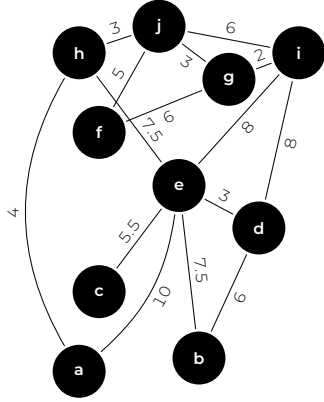


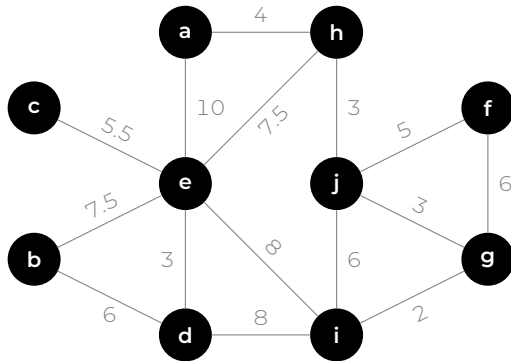


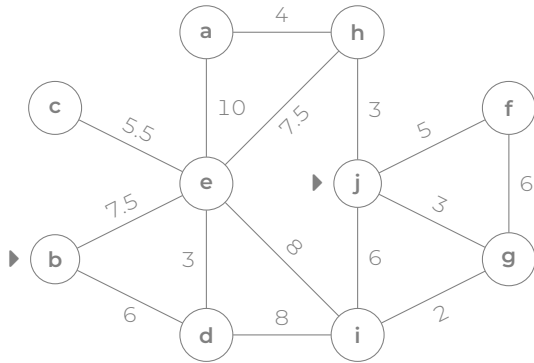


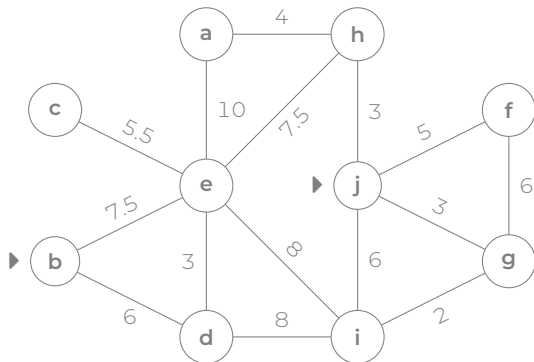






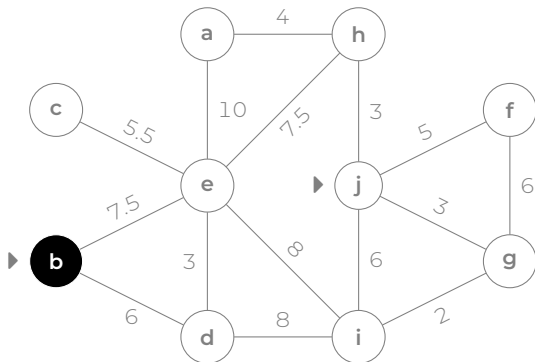




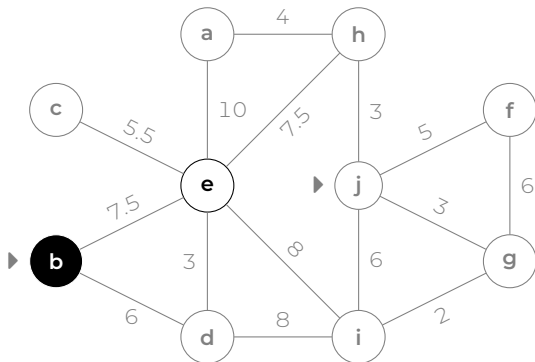


Breitensuche

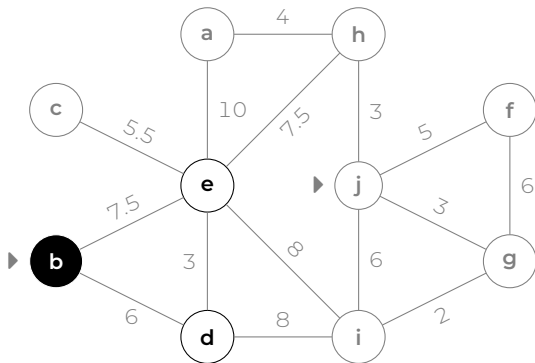
breadth-first search



Breitensuche
breadth-first search



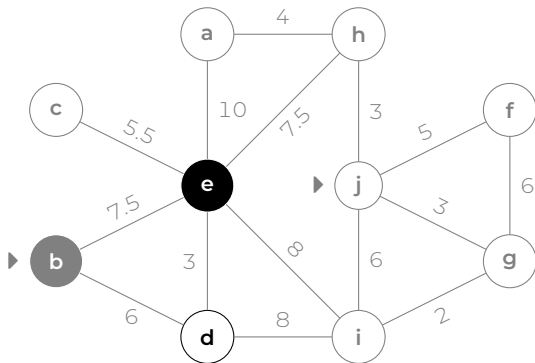
Breitensuche
breadth-first search



Breitensuche

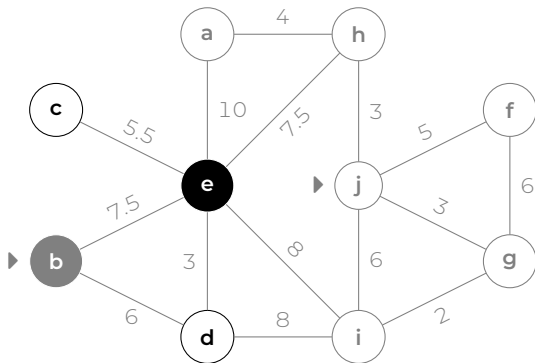
breadth-first search

Warteschlange



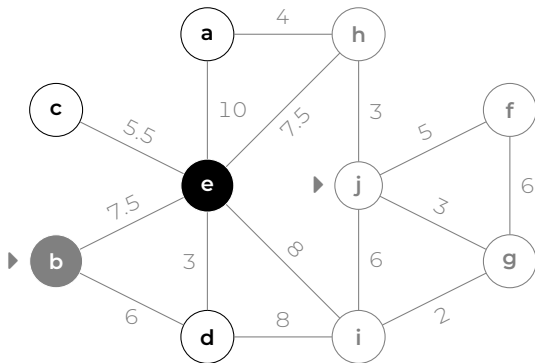
Breitensuche
breadth-first search

Warteschlange



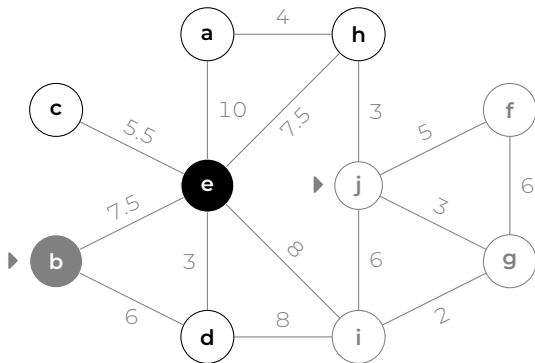
Breitensuche
breadth-first search

Warteschlange



Breitensuche
breadth-first search

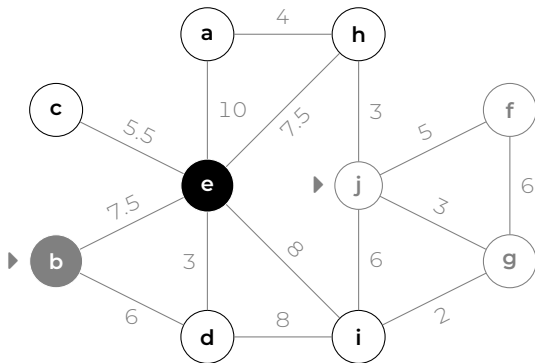
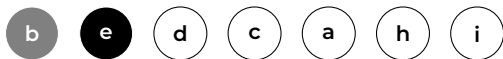
Warteschlange



Breitensuche

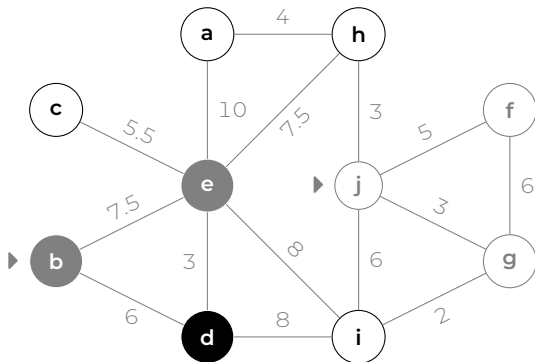
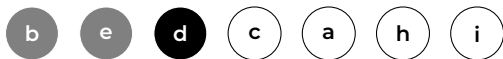
breadth-first search

Warteschlange



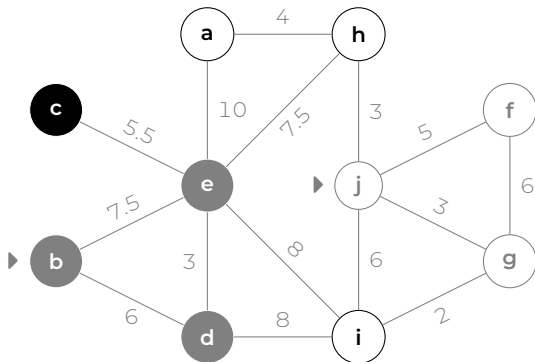
Breitensuche
breadth-first search

Warteschlange



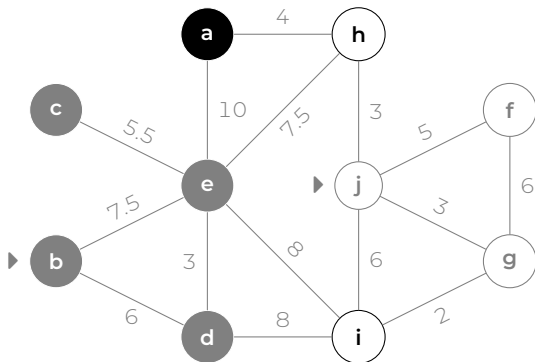
Breitensuche
breadth-first search

Warteschlange



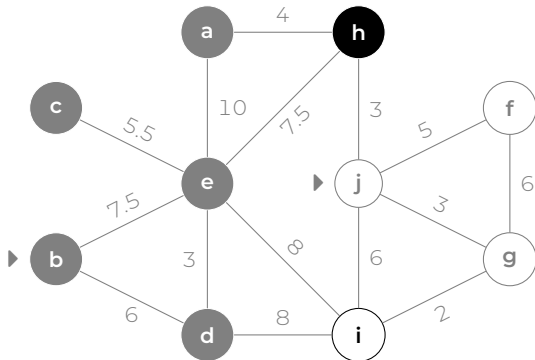
Breitensuche
breadth-first search

Warteschlange



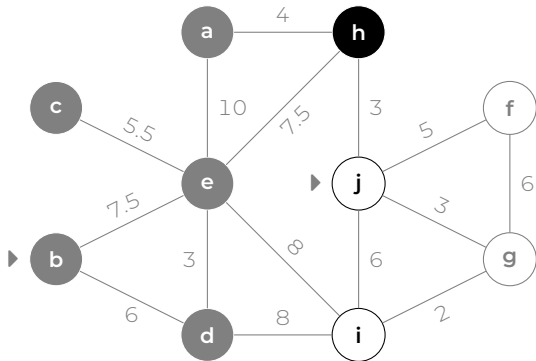
Breitensuche
breadth-first search

Warteschlange



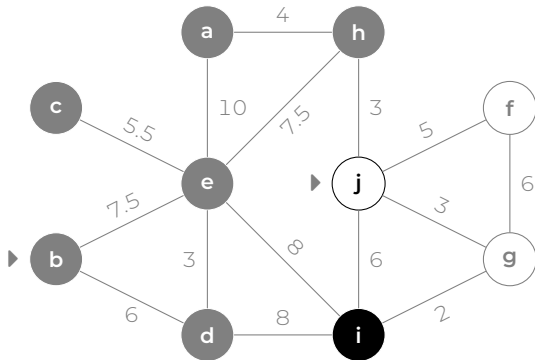
Breitensuche
breadth-first search

Warteschlange



Breitensuche
breadth-first search

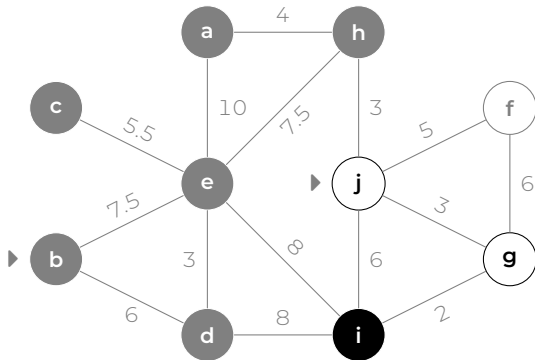
Warteschlange



Breitensuche

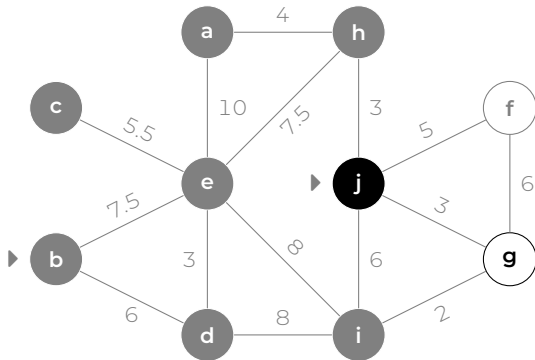
breadth-first search

Warteschlange

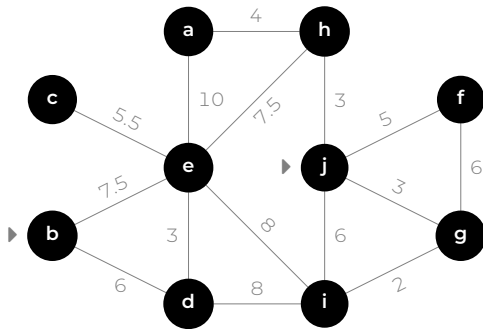


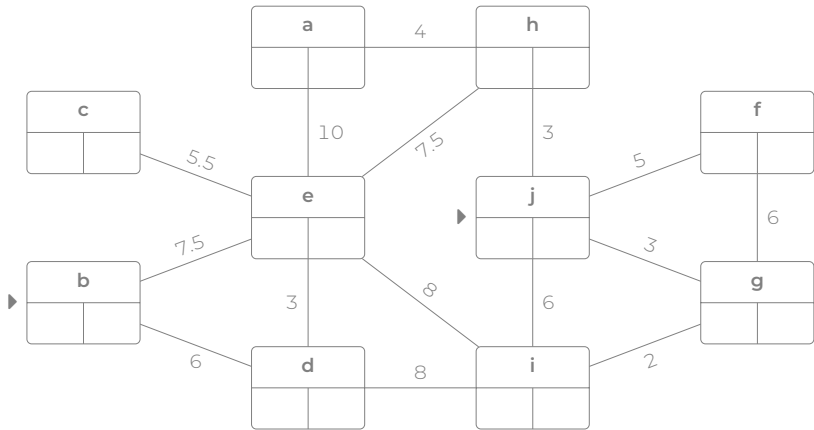
Breitensuche
breadth-first search

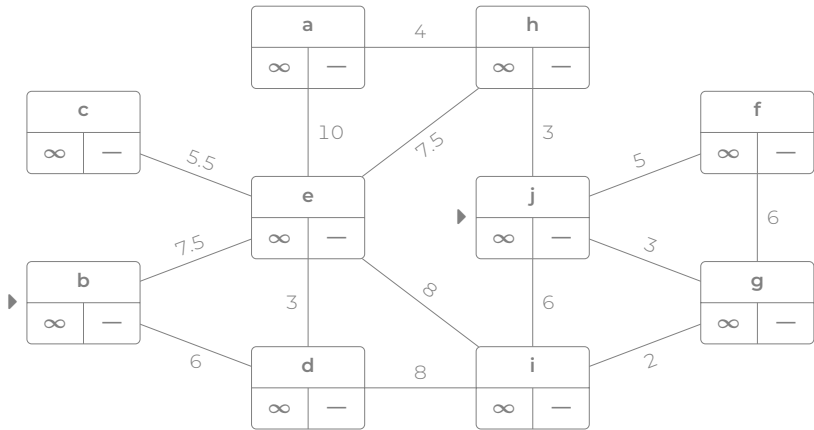
Warteschlange

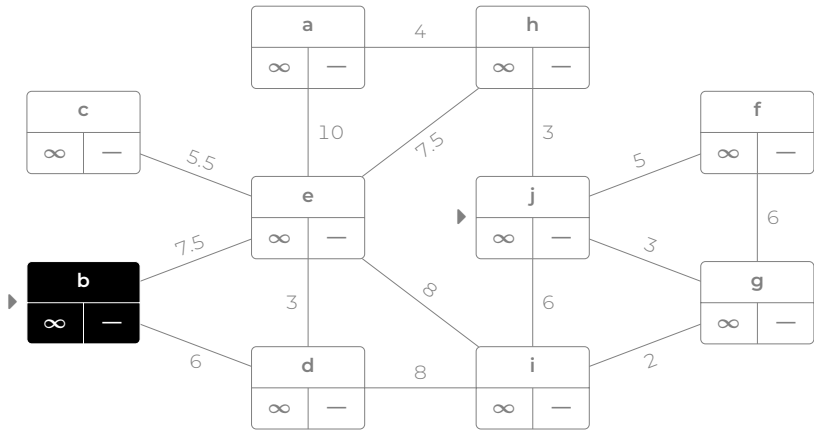


Breitensuche
breadth-first search

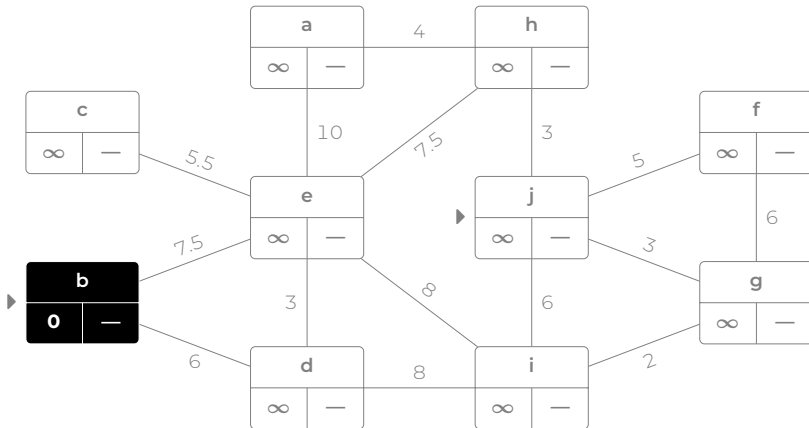






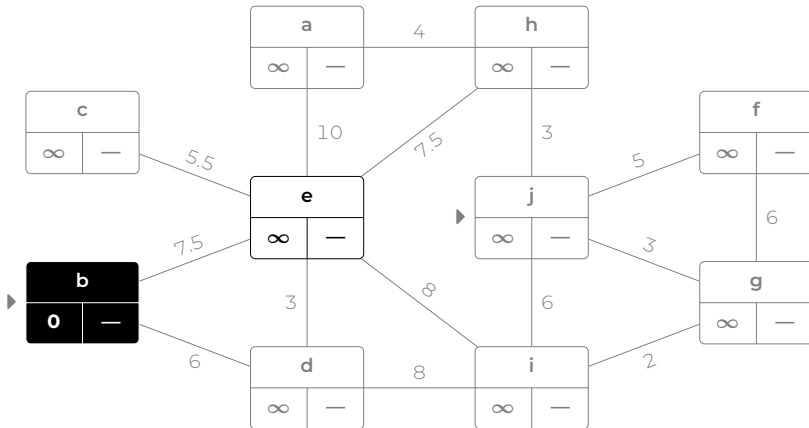


a	∞	b	0	c	∞	d	∞	e	∞	f	∞	g	∞	h	∞	i	∞	j	∞
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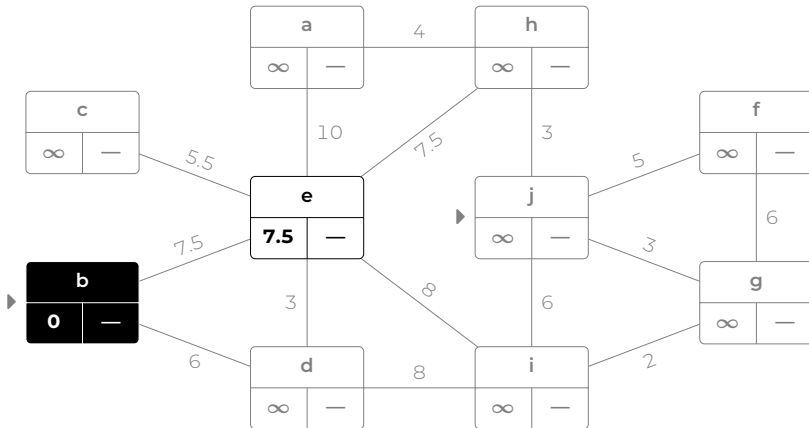
Prioritätswarteschlange

a	∞	b	0	c	∞	d	∞	e	∞	f	∞	g	∞	h	∞	i	∞	j	∞
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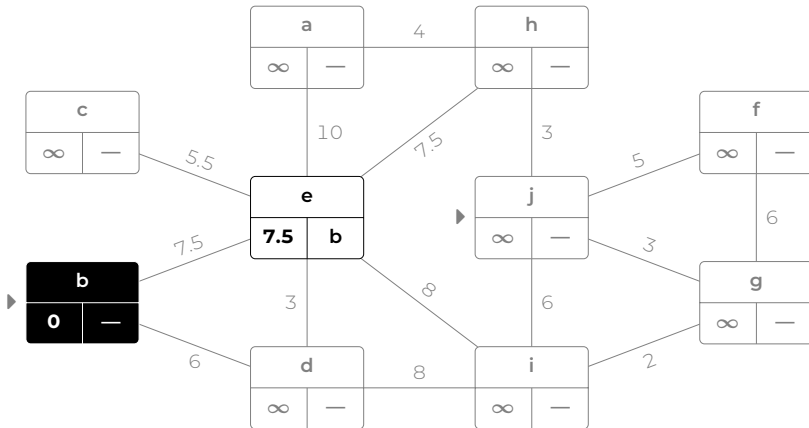
Prioritätswarteschlange

a	∞	b	0	c	∞	d	∞	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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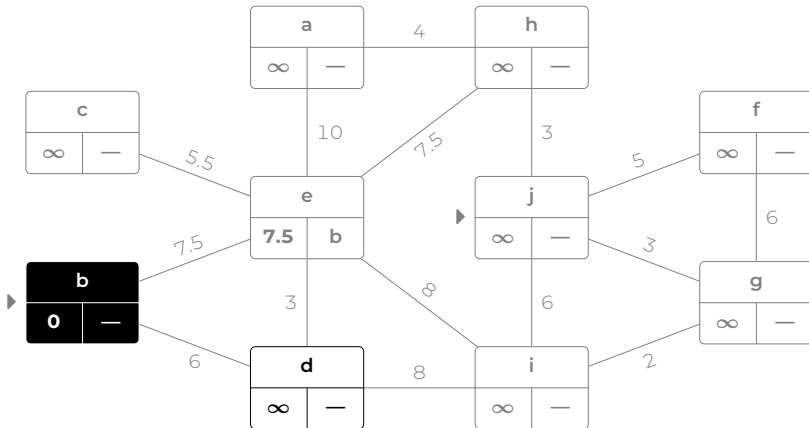
Prioritätswarteschlange

a	∞	b	0	c	∞	d	∞	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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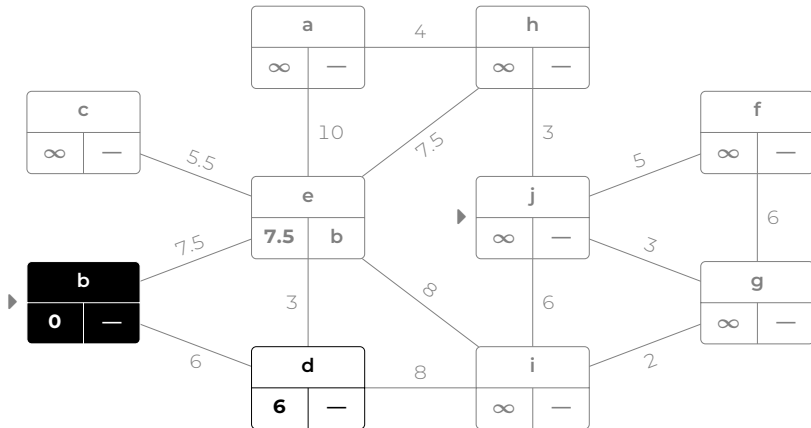
Prioritätswarteschlange

a	∞	b	0	c	∞	d	∞	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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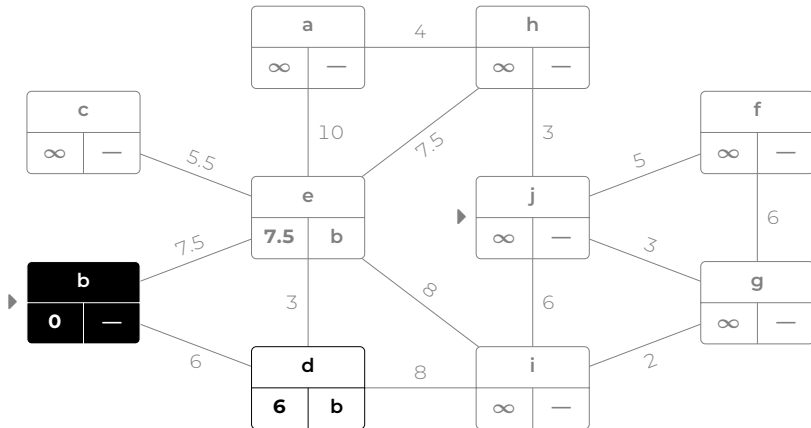
Prioritätswarteschlange

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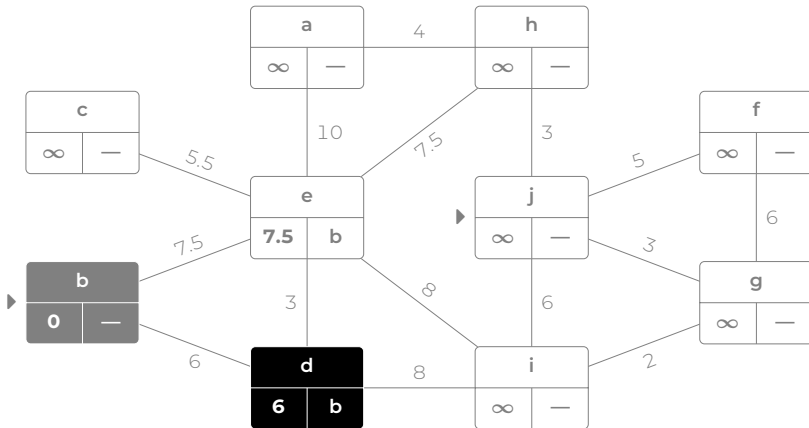
Prioritätswarteschlange

a	∞	b	0	c	∞	d	6	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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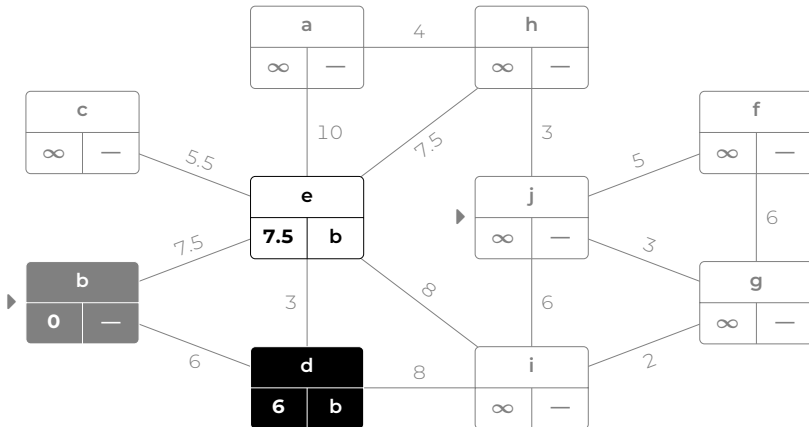
Prioritätswarteschlange

a	∞	b	0	c	∞	d	6	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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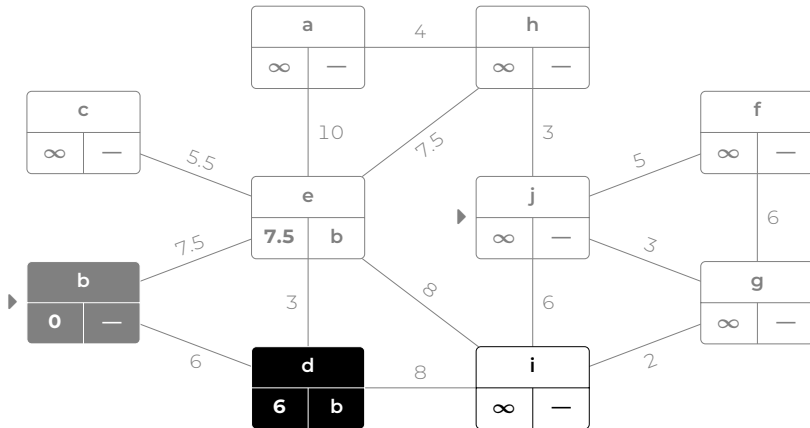
Prioritätswarteschlange

a	∞	b	0	c	∞	d	6	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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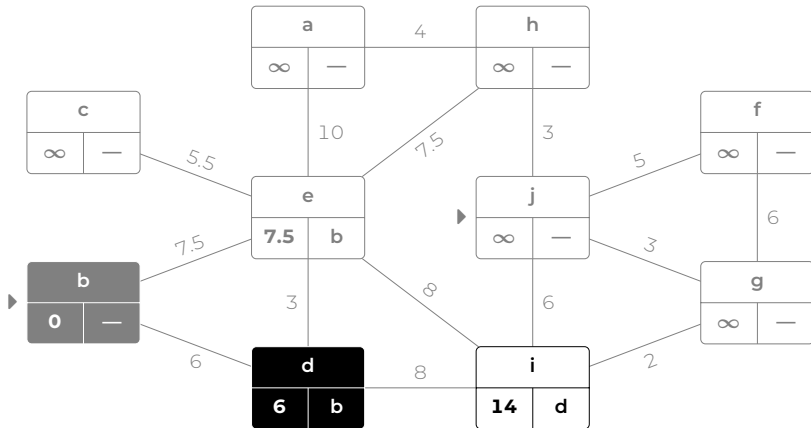
Prioritätswarteschlange

a	∞	b	0	c	∞	d	6	e	7.5	f	∞	g	∞	h	∞	i	∞	j	∞
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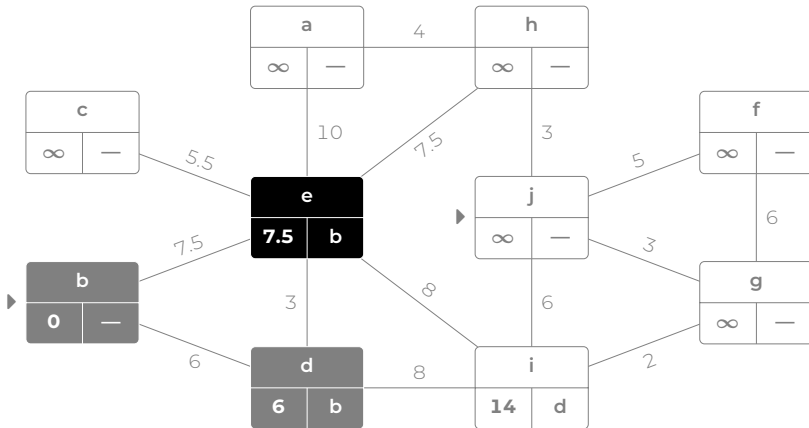
Prioritätswarteschlange

a	∞	b	0	c	∞	d	6	e	7.5	f	∞	g	∞	h	∞	i	14	j	∞
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Prioritätswarteschlange

a	∞	b	0	c	∞	d	6	e	7.5	f	∞	g	∞	h	∞	i	14	j	∞
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Prioritätswarteschlange

a	∞
---	----------

b	0
---	---

c	13
---	----

d	6
---	---

e	7.5
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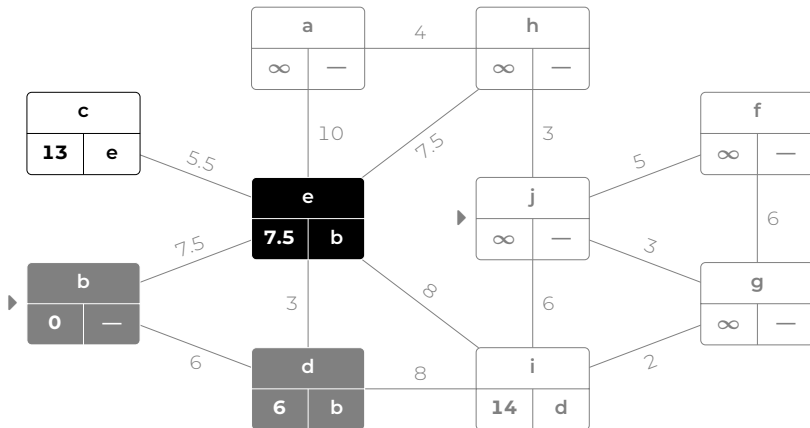
f	∞
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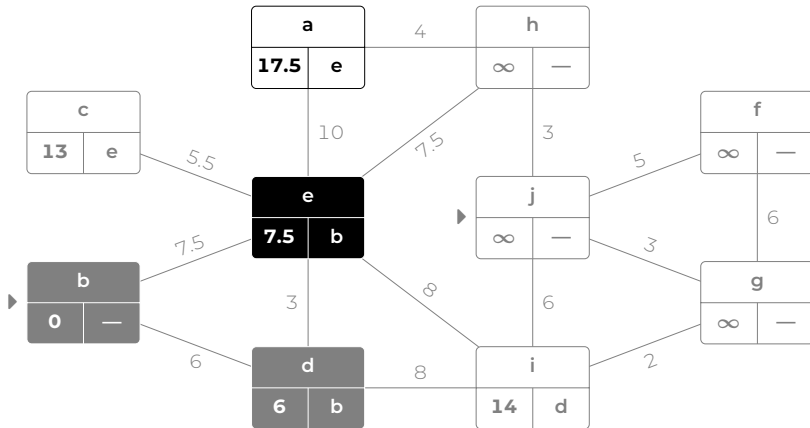
g	∞
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h	∞
---	----------

i	14
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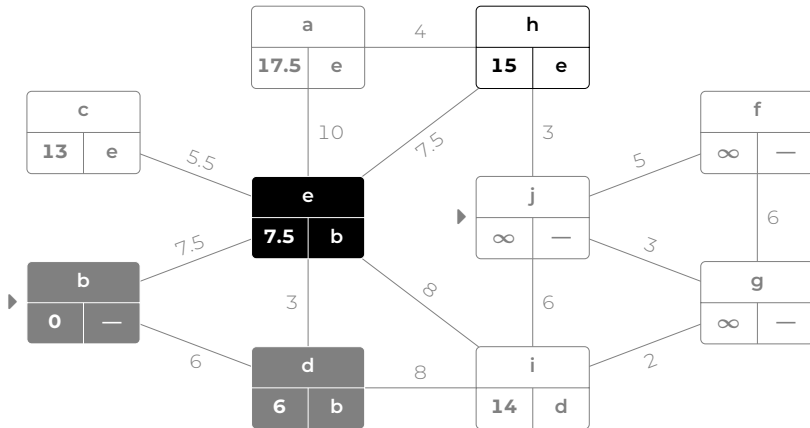
j	∞
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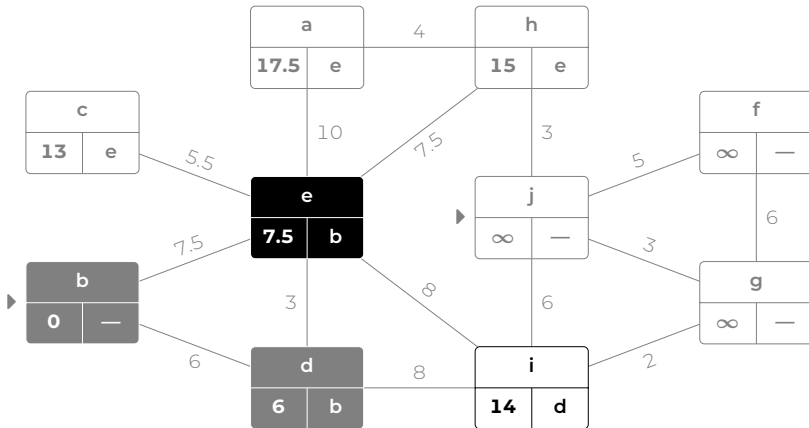
Prioritätswarteschlange

a	17.5	b	0	c	13	d	6	e	7.5	f	∞	g	∞	h	15	i	14	j	∞
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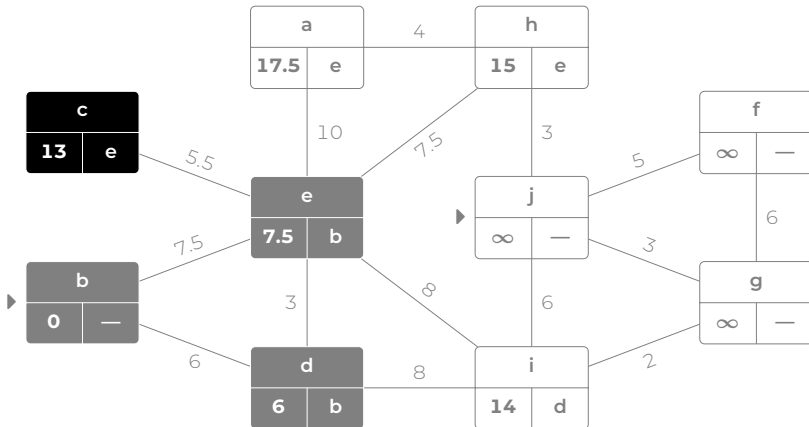
Prioritätswarteschlange

a	17.5	b	0	c	13	d	6	e	7.5	f	∞	g	∞	h	15	i	14	j	∞
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Prioritätswarteschlange

a	17.5	b	0	c	13	d	6	e	7.5	f	∞	g	∞	h	15	i	14	j	∞
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Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

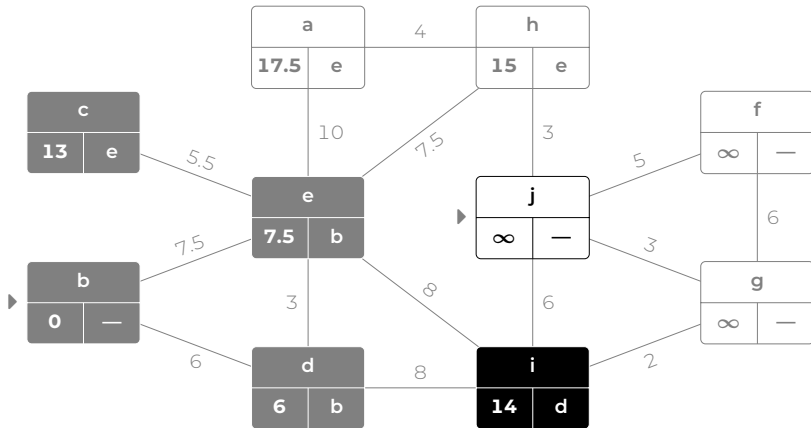
f ∞

g ∞

h 15

i 14

j ∞



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

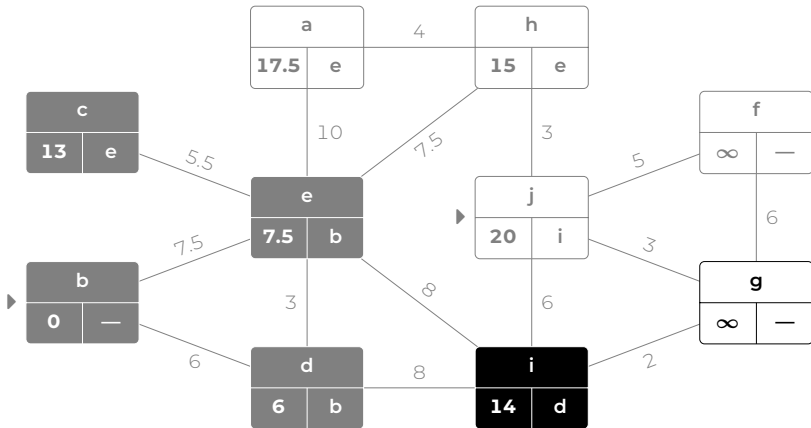
f ∞

g ∞

h 15

i 14

j 20



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

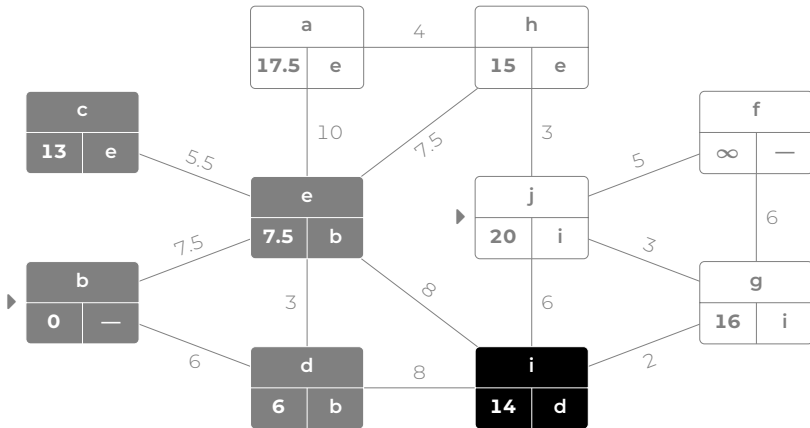
f ∞

g 16

h 15

i 14

j 20



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

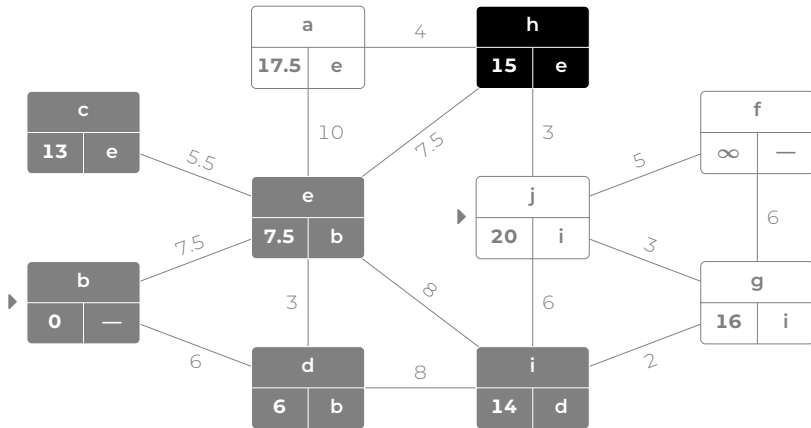
f ∞

g 16

h 15

i 14

j 20



a 17.5

b 0

c 13

d 6

e 7.5

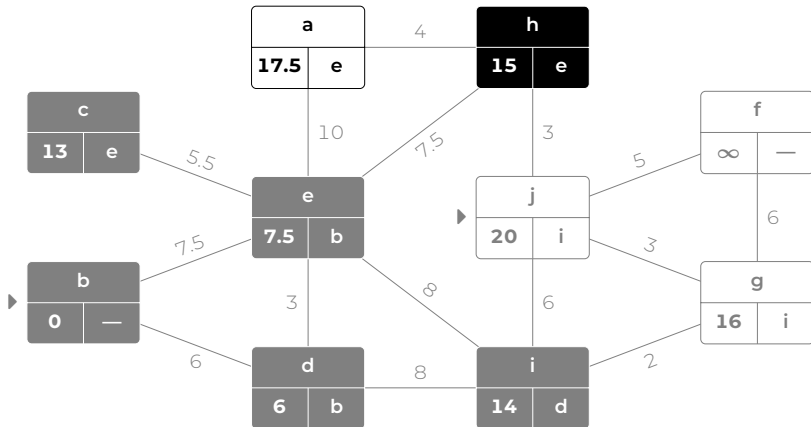
f ∞

g 16

h 15

i 14

j 20



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

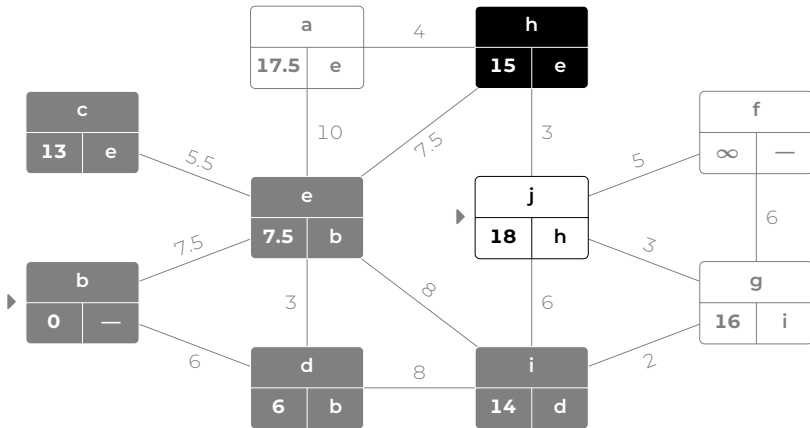
f ∞

g 16

h 15

i 14

j 18



Prioritätswarteschlange

a	17.5
---	------

b	0
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c	13
---	----

d	6
---	---

e	7.5
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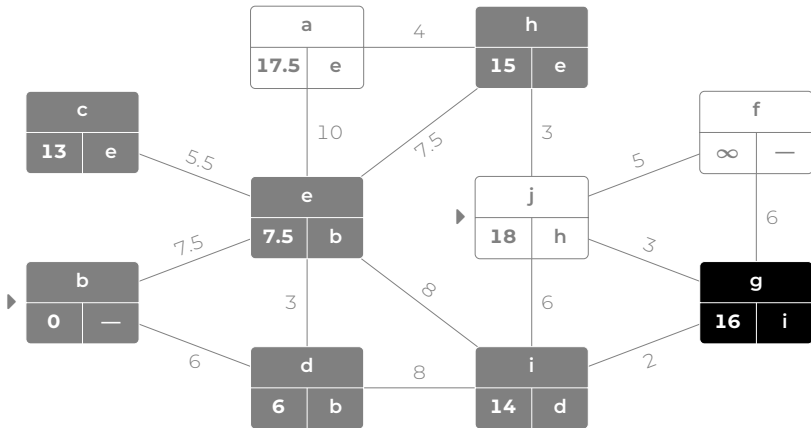
f	∞
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g	16
---	----

h	15
---	----

i	14
---	----

j	18
---	----



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

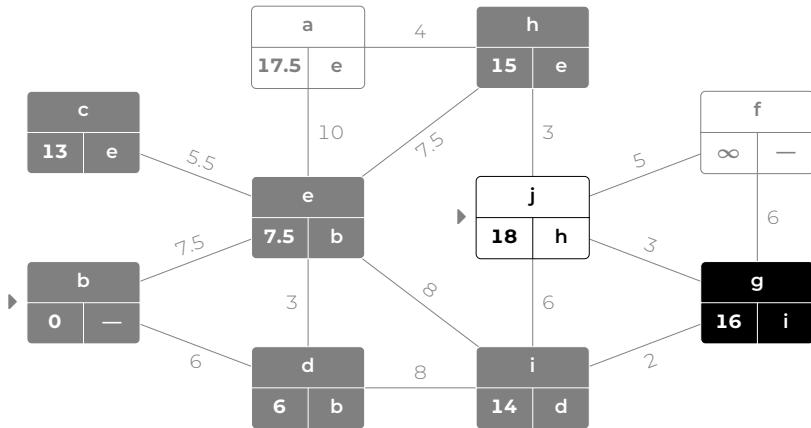
f ∞

g 16

h 15

i 14

j 18



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

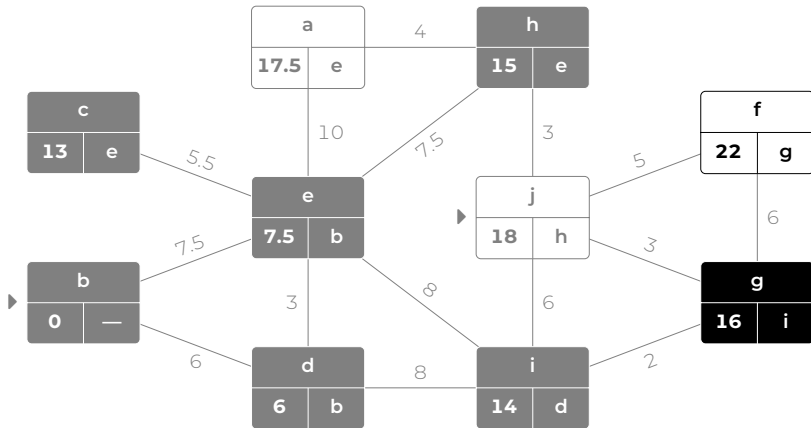
f 22

g 16

h 15

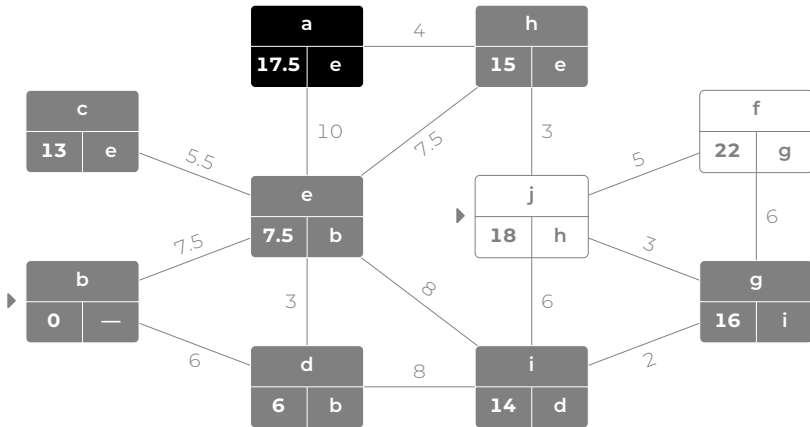
i 14

j 18



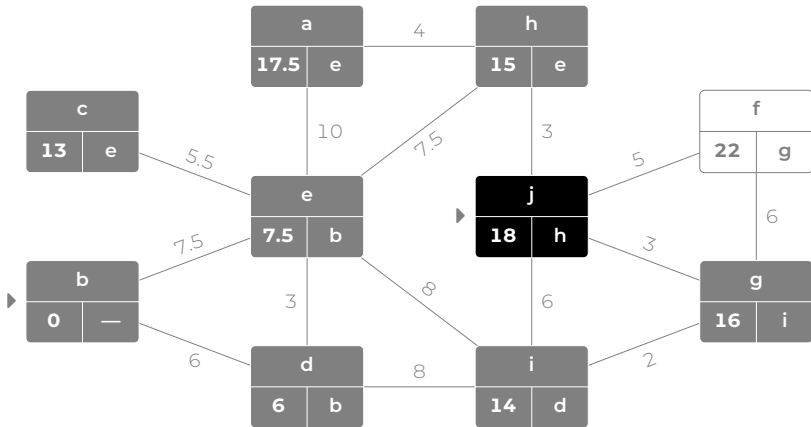
Prioritätswarteschlange

a	17.5	b	0	c	13	d	6	e	7.5	f	22	g	16	h	15	i	14	j	18
---	------	---	---	---	----	---	---	---	-----	---	----	---	----	---	----	---	----	---	----

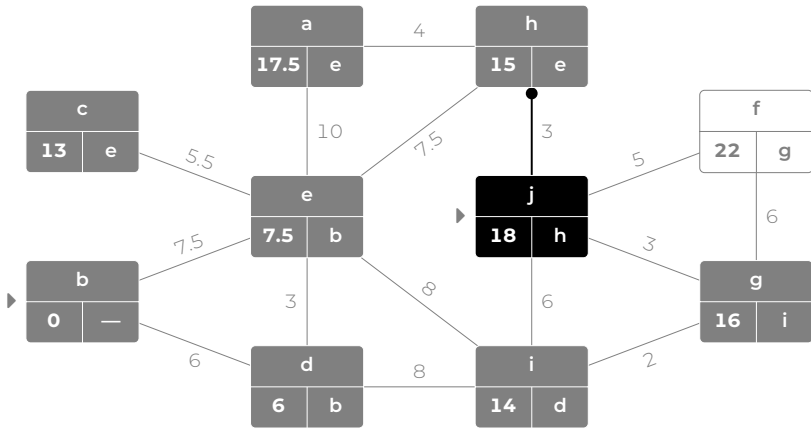


Prioritätswarteschlange

a	17.5	b	0	c	13	d	6	e	7.5	f	22	g	16	h	15	i	14	j	18
---	------	---	---	---	----	---	---	---	-----	---	----	---	----	---	----	---	----	---	----



a	17.5	b	0	c	13	d	6	e	7.5	f	22	g	16	h	15	i	14	j	18
---	------	---	---	---	----	---	---	---	-----	---	----	---	----	---	----	---	----	---	----



Prioritätswarteschlange

a 17.5

b 0

c 13

d 6

e 7.5

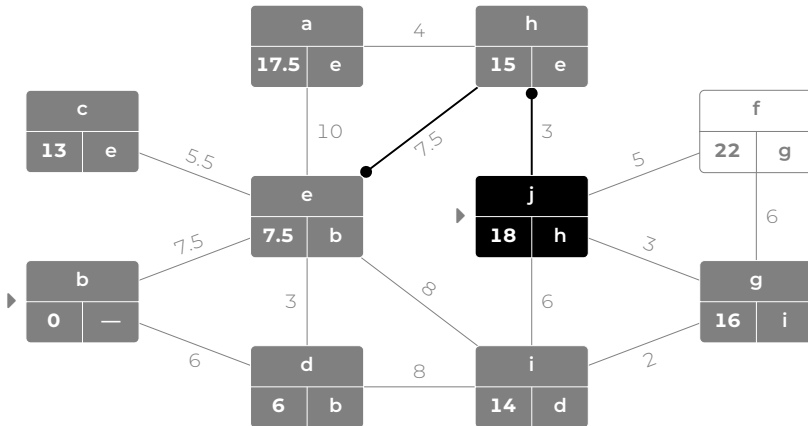
f 22

g 16

h 15

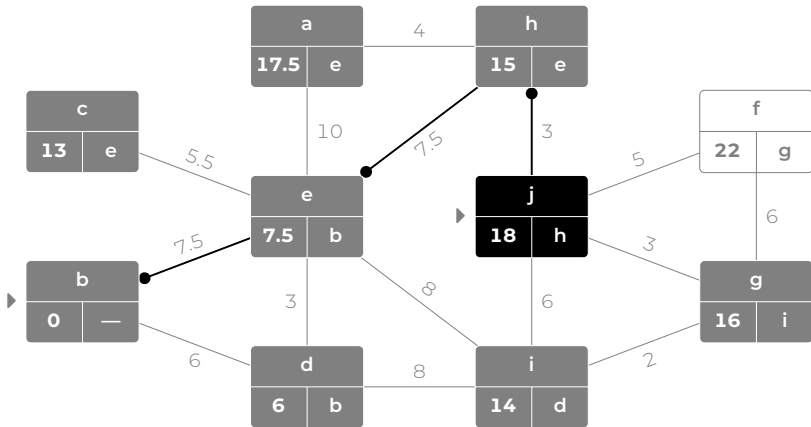
i 14

j 18

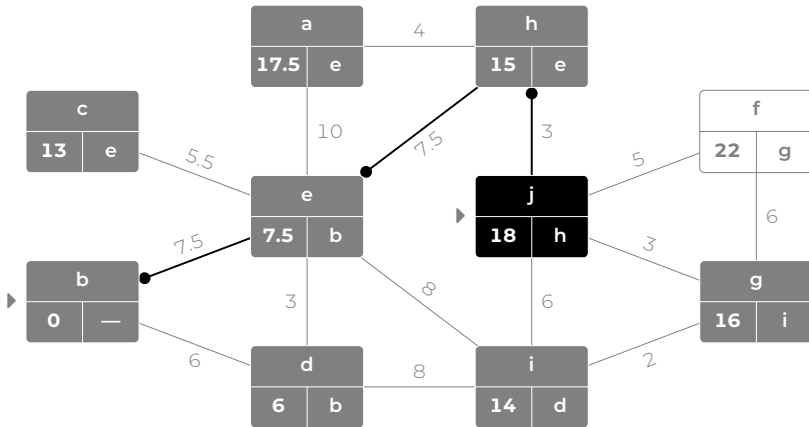


Prioritätswarteschlange

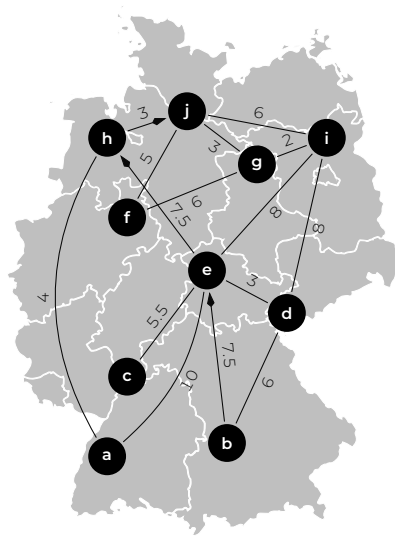
a	17.5	b	0	c	13	d	6	e	7.5	f	22	g	16	h	15	i	14	j	18
---	------	---	---	---	----	---	---	---	-----	---	----	---	----	---	----	---	----	---	----



a	17.5	b	0	c	13	d	6	e	7.5	f	22	g	16	h	15	i	14	j	18
---	------	---	---	---	----	---	---	---	-----	---	----	---	----	---	----	---	----	---	----



Dijkstra-Algorithmus



Eingabe: Graph $G = (V, E)$, Start- und Endknoten $s, e \subseteq V$

Erzeuge Prioritätswarteschlange Q

Für alle $v \in V$ **tue:**

$v.\text{Kosten} \leftarrow \infty$

$v.\text{Vorgänger} \leftarrow ?$

Füge v zu Q hinzu

$s.\text{Kosten} \leftarrow 0$

Solange Q nicht leer ist, **tue:**

$u \leftarrow$ Wähle v mit minimalen Kosten aus Q

Wenn u das gesuchte e ist, **dann:**

Gib Pfad von e über Vorgänger **zurück**

Entferne u aus Q

Für alle Nachbarn v von u **tue:**

$a \leftarrow u.\text{Kosten} + \text{Kosten der Kante } \{u, v\}$

Wenn $a < v.\text{Kosten}$, **dann:**

$v.\text{Kosten} \leftarrow a$

$v.\text{Vorgänger} \leftarrow u$

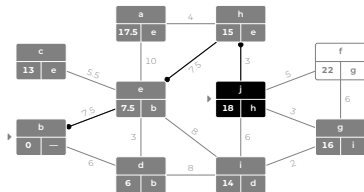
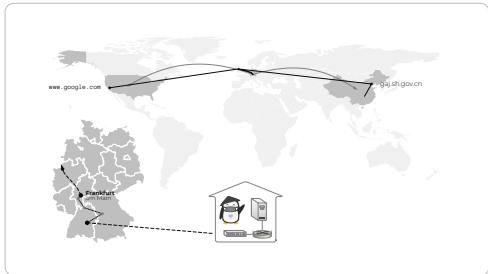
Gib zurück, dass kein Pfad gefunden wurde

- Korrektheitsbeweise

- Korrektheitsbeweise
- Laufzeitanalysen

- Korrektheitsbeweise
- Laufzeitanalysen
- Optimierungen und alternative Ansätze

- Korrektheitsbeweise
- Laufzeitanalysen
- Optimierungen und alternative Ansätze
- Implementierung



Eingabe: Graph $G = (V, E)$, Start- und Endknoten $s, e \in V$

Erzeuge Prioritätswarteschlange Q

Für alle $v \in V$ **tue:**

$v.\text{Kosten} \leftarrow \infty$

$v.\text{Vorgänger} \leftarrow ?$

Füge v zu Q hinzu

$s.\text{Kosten} \leftarrow 0$

Solange Q nicht leer ist, **tue:**

$u \leftarrow$ Wähle v mit minimalen Kosten aus Q

Wenn u das gesuchte e ist, **dann:**

Gib Pfad von e über Vorgänger **zurück**

Entferne u aus Q

Für alle Nachbarn v von u **tue:**

$\alpha \leftarrow u.\text{Kosten} + \text{Kosten der Kante } \{u, v\}$

Wenn $\alpha < v.\text{Kosten}$, **dann:**

$v.\text{Kosten} \leftarrow \alpha$

$v.\text{Vorgänger} \leftarrow u$

Gib zurück, dass kein Pfad gefunden wurde



VON
PHYSIK,
INFORMATIK
UND DEM GANZEN REST

205.770

205.770
19.973



Alice



Alice

Hallo



Bob



Alice

Hallo



Bob



Eve



Alice



Bob

Hallo

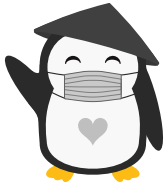


Alice



Bob

`encrypt(Hallo, 7)`



Alice



Bob

`encrypt(Hallo, 7)`



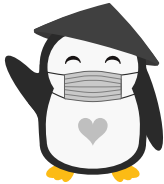
Alice

Ohssv



Bob

`encrypt(Hallo, 7)`



Alice

Ohssv

`decrypt(Ohssv, 7)`



Bob

encrypt(Hallo, 7)



Alice

Ohssv

decrypt(Ohssv, 7)



Bob

encrypt(Hallo, 7)



Alice

Ohssv

Symmetrisch

decrypt(Ohssv, 7)



Bob

encrypt(Hallo, 7)



Alice

decrypt(Ohssv, 7)



Bob

Ohssv

Symmetrisch

encrypt(Hallo, 7)



Alice

Ohssv

Symmetrisch

decrypt(Ohssv, 7)



Bob

encrypt(Hallo, 7)



Alice

Ohssv

Asymmetrisch

decrypt(Ohssv, 4)



Bob



$$p = 11 \quad g = 7$$



$$p = 11 \quad g = 7$$

$$a = 3$$



$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$



$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$



$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$

$$b = 2$$



$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$

$$b = 2$$



$$g^b \bmod p$$

$$7^2 \bmod 11 = \mathbf{5}$$

$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$

$$g^a \bmod p = 2$$

$$g^b \bmod p = 5$$

$$7^2 \bmod 11 = \mathbf{5}$$

$$b = 2$$



$$g^b \bmod p$$

$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$

$$5^3 \bmod 11 = \mathbf{4}$$

$$g^a \bmod p = 2$$

$$g^b \bmod p = 5$$

$$b = 2$$



$$g^b \bmod p$$

$$7^2 \bmod 11 = \mathbf{5}$$

$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$

$$5^3 \bmod 11 = \mathbf{4}$$

$$g^a \bmod p = 2$$

$$g^b \bmod p = 5$$

$$g^b \bmod p$$

$$7^2 \bmod 11 = \mathbf{5}$$

$$2^2 \bmod 11 = \mathbf{4}$$

$$b = 2$$



$$p = 11 \quad g = 7$$

$$a = 3$$



$$g^a \bmod p$$

$$7^3 \bmod 11 = \mathbf{2}$$

$$5^3 \bmod 11 = \mathbf{4}$$

$$g^a \bmod p = 2$$

$$g^b \bmod p = 5$$

$$g^b \bmod p$$

$$7^2 \bmod 11 = \mathbf{5}$$

$$2^2 \bmod 11 = \mathbf{4}$$

$$b = 2$$



Diffie-Hellman Schlüsselaustausch

Faktorisierungsproblem

Faktorisierungsproblem



0 / 1

Bit

0 / 1

Bit

0 / 1

Qubit

$|0\rangle$ / $|1\rangle$

Bit

0 / 1

Qubit

$|0\rangle$ / $|\psi\rangle$ / $|1\rangle$
Superposition

$$|\psi\rangle = a \cdot |\mathbf{0}\rangle + b \cdot |\mathbf{1}\rangle$$

$$|\psi\rangle = a \cdot |\mathbf{0}\rangle + b \cdot |\mathbf{1}\rangle$$

$$|\psi\rangle = a \cdot |\mathbf{00}\rangle + b \cdot |\mathbf{11}\rangle$$

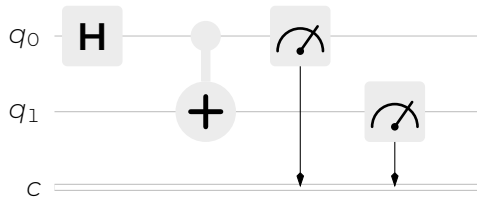
$$|\psi\rangle = a \cdot |\mathbf{0}\rangle + b \cdot |\mathbf{1}\rangle$$

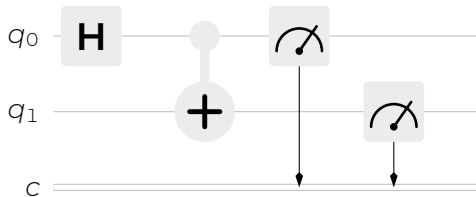
$$|\psi\rangle = a \cdot |\mathbf{10}\rangle + b \cdot |\mathbf{01}\rangle$$

$$|\psi\rangle = a \cdot |\mathbf{0}\rangle + b \cdot |\mathbf{1}\rangle$$

$$|\psi\rangle = a \cdot |\mathbf{10}\rangle + b \cdot |\mathbf{01}\rangle$$

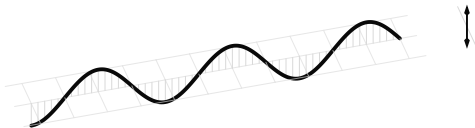
Verschränkung
entanglement



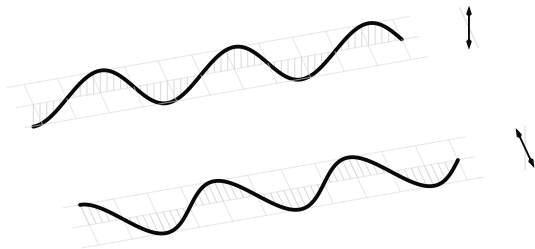


Shor-Algorithmus

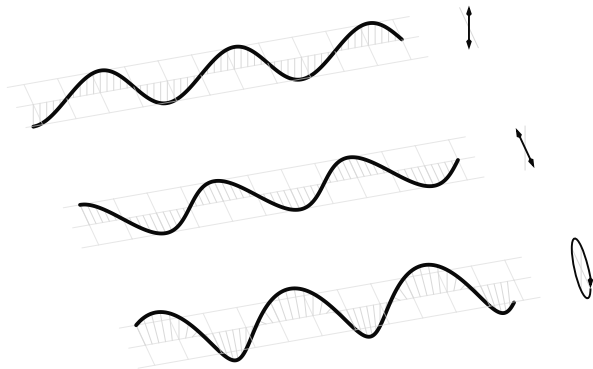
Polarisation



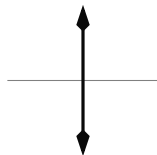
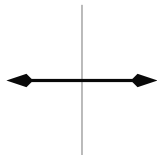
Polarisation

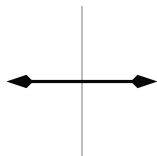


Polarisation

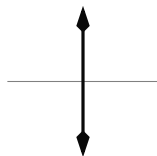


Polarisation

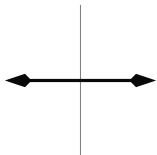




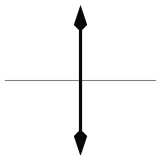
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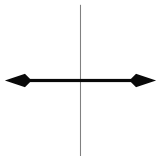
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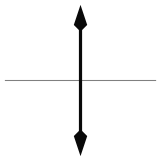
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0



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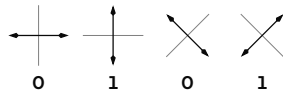


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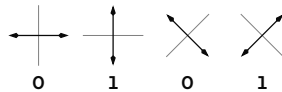




Alice



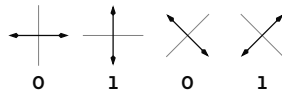
Bob



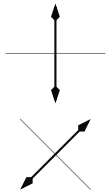
Alice



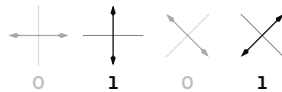
Bob



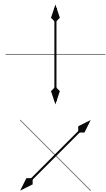
Alice



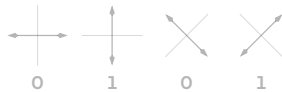
Bob



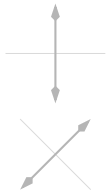
Alice



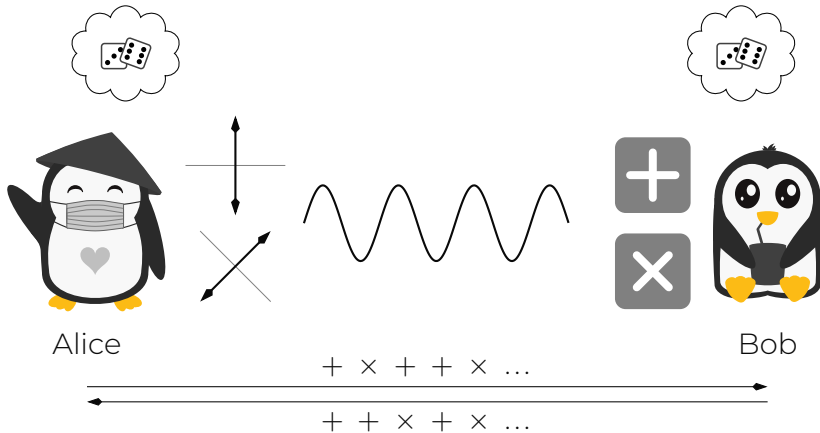
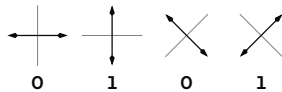
Bob

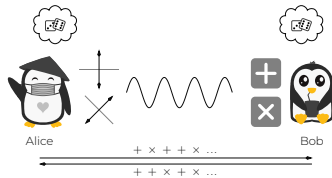
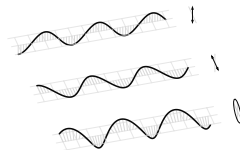
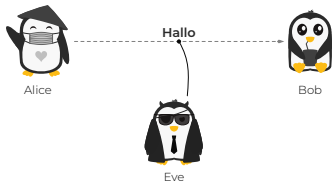


Alice



Bob







VON
**PHYSIK,
INFORMATIK**
UND DEM GANZEN REST

- Transistoren und Mikroelektronik
- Rekursion
- Berechenbarkeits-Modelle
- Rechnerarchitektur
- Verteilte Systeme
- Programmierparadigmen
- Algorithmen und Datenstrukturen
- Angewandte Gebiete der Informatik
- ...













F. Sihler

florian.sihler@uni-ulm.de